

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Text Preprocessing	Cleans, tokenizes, and normalizes student input text before emotion prediction.	Preprocessing Module	Done	High
2	BiLSTM Emotion Detection	Predicts student emotions using a Bidirectional LSTM deep learning model.	BiLSTM Model	Done	High
3	BERT Emotion Classification	Performs contextual emotion detection using a fine-tuned BERT model.	BERT Model	Done	High
4	Mixed Emotion Detection	Identifies primary and secondary emotions using confidence scores.	Emotion Prediction Engine	Done	Medium
5	AI Learning Guidance	Generates personalized learning guidance using Google Gemini AI.	Gemini API Integration	Done	High
6	Emotion Analytics Dashboard	Displays detected emotions, confidence scores, and visual charts.	Streamlit Dashboard	Done	Medium
7	Session History Management	Stores emotion records and AI-generated responses for future reference.	CSV Storage	Done	Medium
8	Interactive User Interface	Provides an easy-to-use Streamlit interface for emotion analysis and personalized learning support.	Streamlit Frontend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Streamlit Dashboard, Interactive User Interface
2	Backend / Logic	3	Text Preprocessing, Emotion Prediction, Mixed Emotion Detection
3	Database / Storage	1	CSV Session History and Emotion Records
4	AI / Integration	2	BERT Model, Google Gemini API
5	Security / Configuration	1	Secure API Key Management using .env